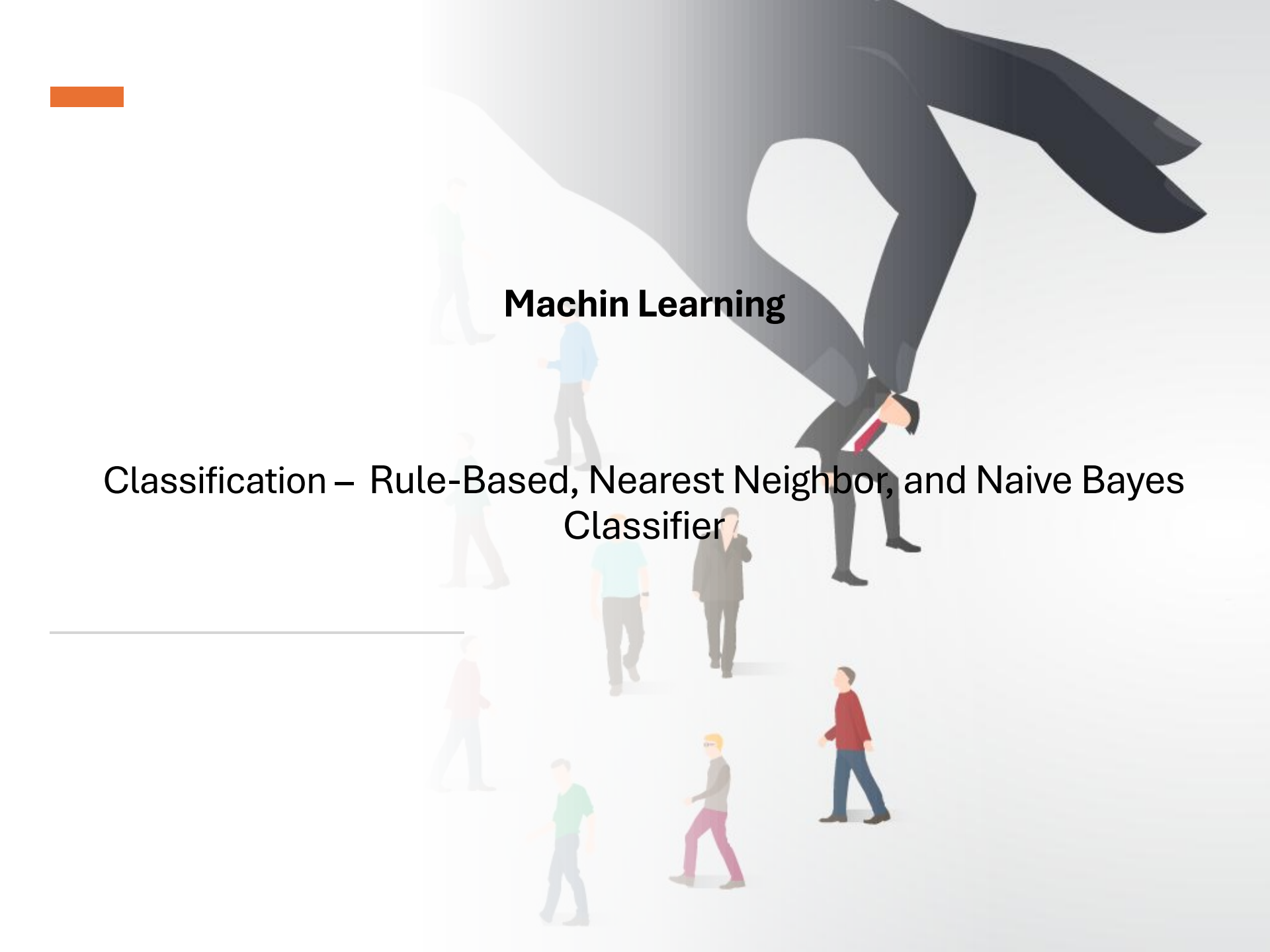


An illustration of a giant, dark grey hand reaching down from the top right corner, holding a man in a black suit and red tie. Several smaller, stylized human figures in various colors (green, blue, grey, teal, brown, red, purple) are walking around the base of the hand. The background is a light grey gradient.

Machin Learning

Classification – Rule-Based, Nearest Neighbor, and Naive Bayes Classifier



Topics

- **Rule-Based Classifier**
- Nearest Neighbor Classifier
- Naive Bayes Classifier

Rule-Based Classifier

- Classify records by using a collection of “if...then...” rules
- Rule: $(Condition) \rightarrow y$
 - Condition is a conjunctions of attributes called LHS, antecedent or condition
 - y is the class label called RHS or consequent
- Examples of classification rules for an animal dataset:
 - $(Blood\ Type = Warm) \wedge (Lay\ Eggs = Yes) \rightarrow Birds$
 - $(Taxable\ Income < 50K) \wedge (Refund = Yes) \rightarrow Evade = No$

Using a Rule-Based Classifier

A rule R **covers** an instance x if the attributes of the instance satisfy the condition of the rule. Such a rule can be used for classification.

R1: (Give Birth = no) $\wedge$ (Can Fly = yes) $\rightarrow$ Birds

R2: (Give Birth = no) $\wedge$ (Live in Water = yes) $\rightarrow$ Fishes

R3: (Give Birth = yes) $\wedge$ (Blood Type = warm) $\rightarrow$ Mammals

R4: (Give Birth = no) $\wedge$ (Can Fly = no) $\rightarrow$ Reptiles

R5: (Live in Water = sometimes) $\rightarrow$ Amphibians

Rule base

Name	Blood Type	Give Birth	Can Fly	Live in Water	Class
hawk	warm	no	yes	no	?
grizzly bear	warm	yes	no	no	?

The rule R1 covers: *hawk* $\rightarrow$ *Bird*

The rule R3 covers: *grizzly bear* $\rightarrow$ *Mammal*

Ordered Rule Set vs. Voting

- Rules are rank ordered according to their priority
 - An ordered rule set is known as a decision list
- When a test record is presented to the classifier
 - It is assigned to the class label of the highest ranked rule it has triggered (R3 is selected below -> Amphibians)
 - If none of the rules fired, it is assigned to the default class

R1: (Give Birth = no) $\wedge$ (Can Fly = yes) $\rightarrow$ Birds

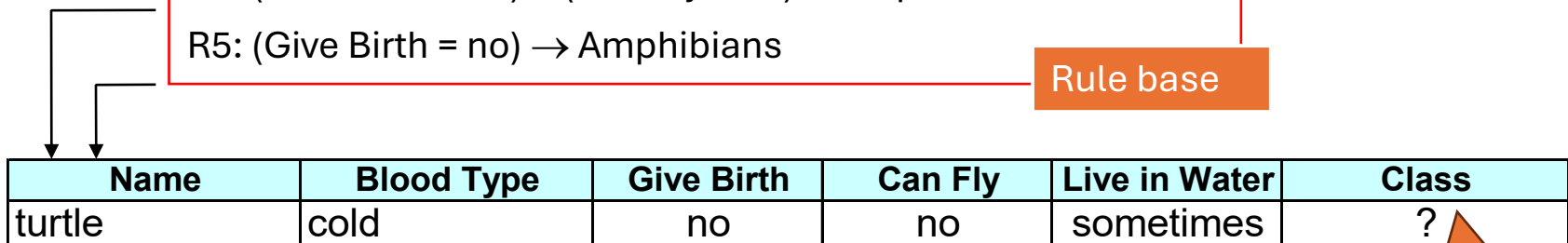
R2: (Give Birth = yes) $\wedge$ (Blood Type = warm) $\rightarrow$ Mammals

R3: (Live in Water = sometimes) $\rightarrow$ Amphibians

R4: (Give Birth = no) $\wedge$ (Can Fly = no) $\rightarrow$ Reptiles

R5: (Give Birth = no) $\rightarrow$ Amphibians

Rule base

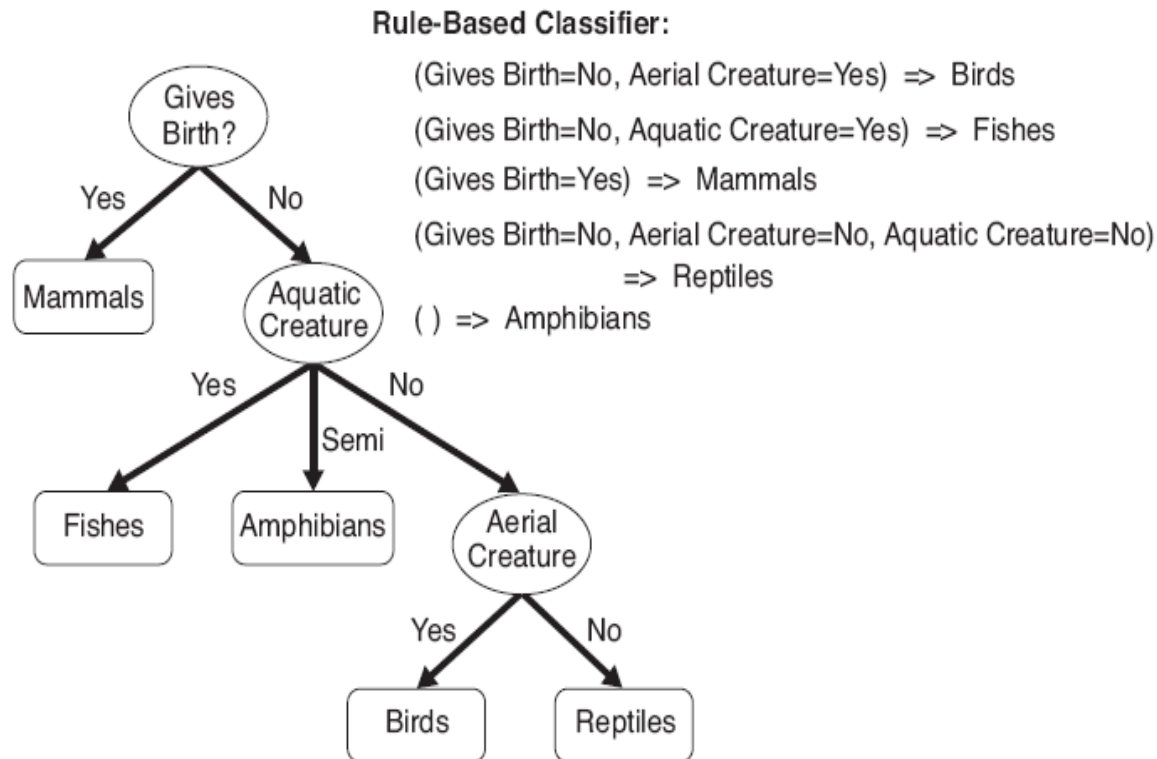


Name	Blood Type	Give Birth	Can Fly	Live in Water	Class
turtle	cold	no	no	sometimes	?

- Alternative: (weighted) voting by all matching rules (-> Amphibians)

R3, 4 and 5
cover the
observation

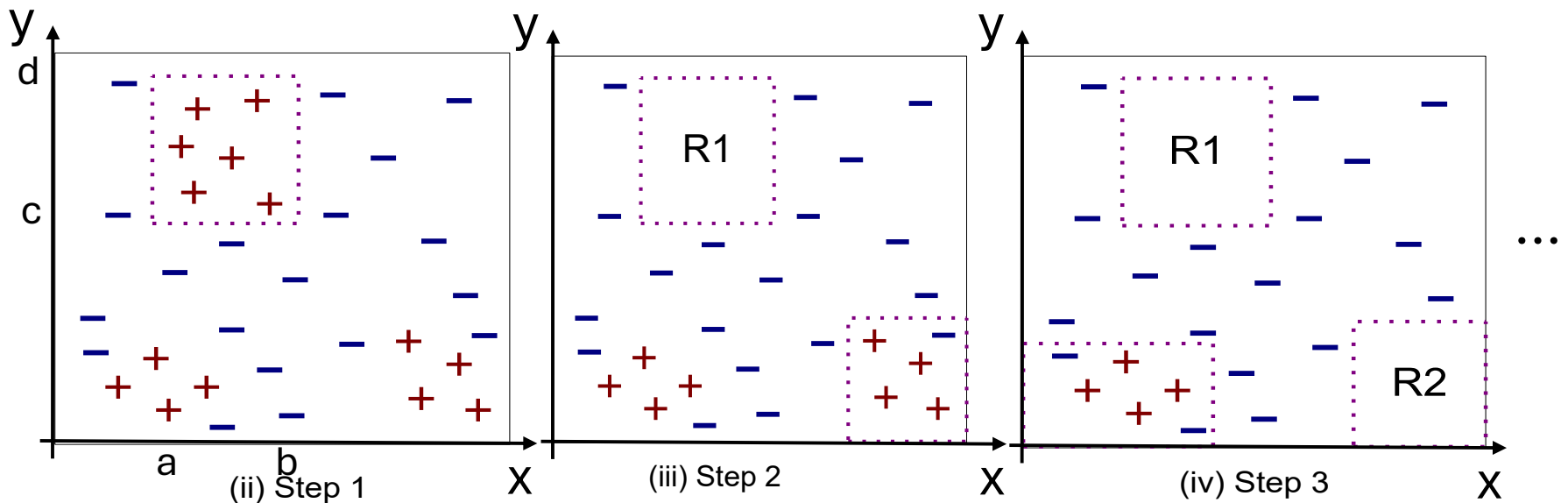
Rules From Decision Trees



- Rules are created by reading the decisions in tree branches from the root to a final node.
- Rule set contains as much information as the tree.
- Rules can be simplified (similar to pruning of the tree).
- Example: C4.5rules

Direct Methods of Rule Generation

- Extract rules directly from the data.
- Sequential Covering (Example: try to cover class +)



$$R1: a > x > b \wedge c > y > d \rightarrow \text{class } +$$

Advantages of Rule-Based Classifiers

As expressive
as decision
trees

Easy to
interpret

Easy to
generate

Can classify
new instances
rapidly

Performance
comparable to
decision trees

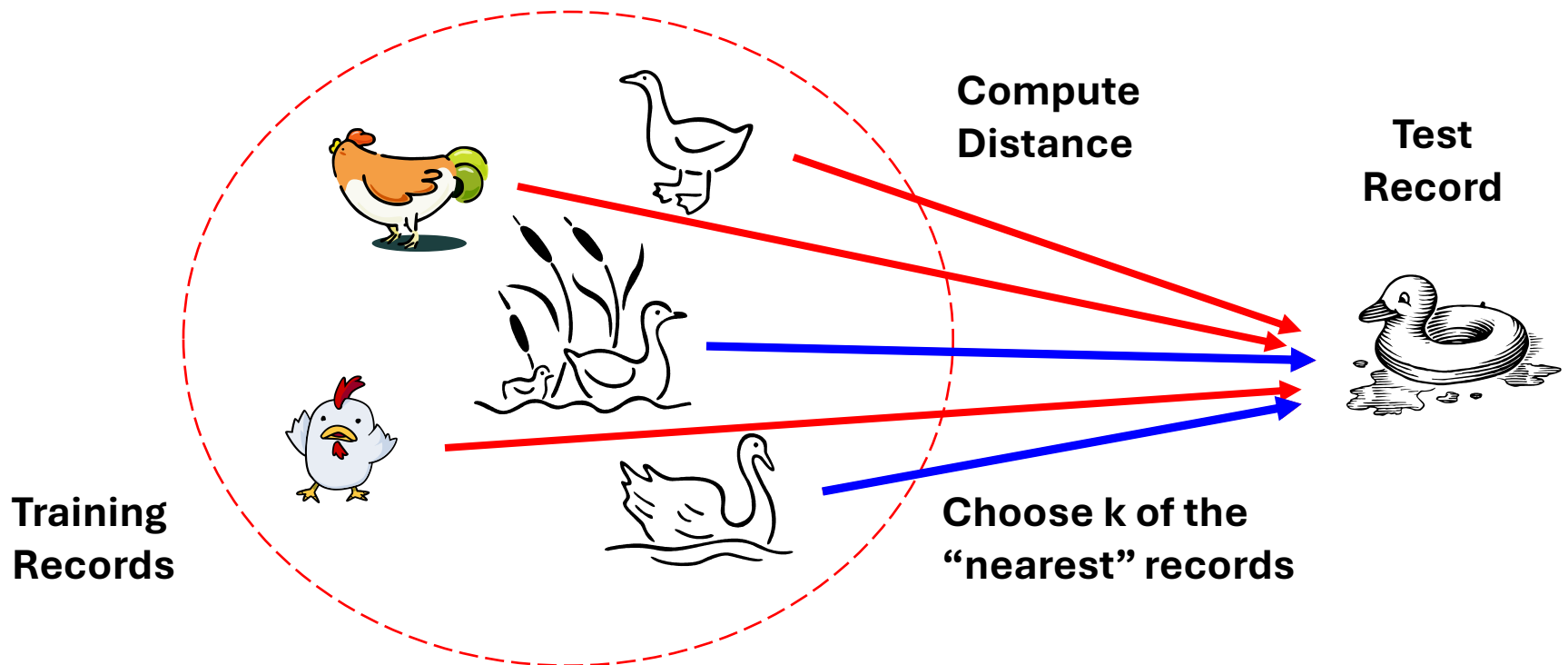


Topics

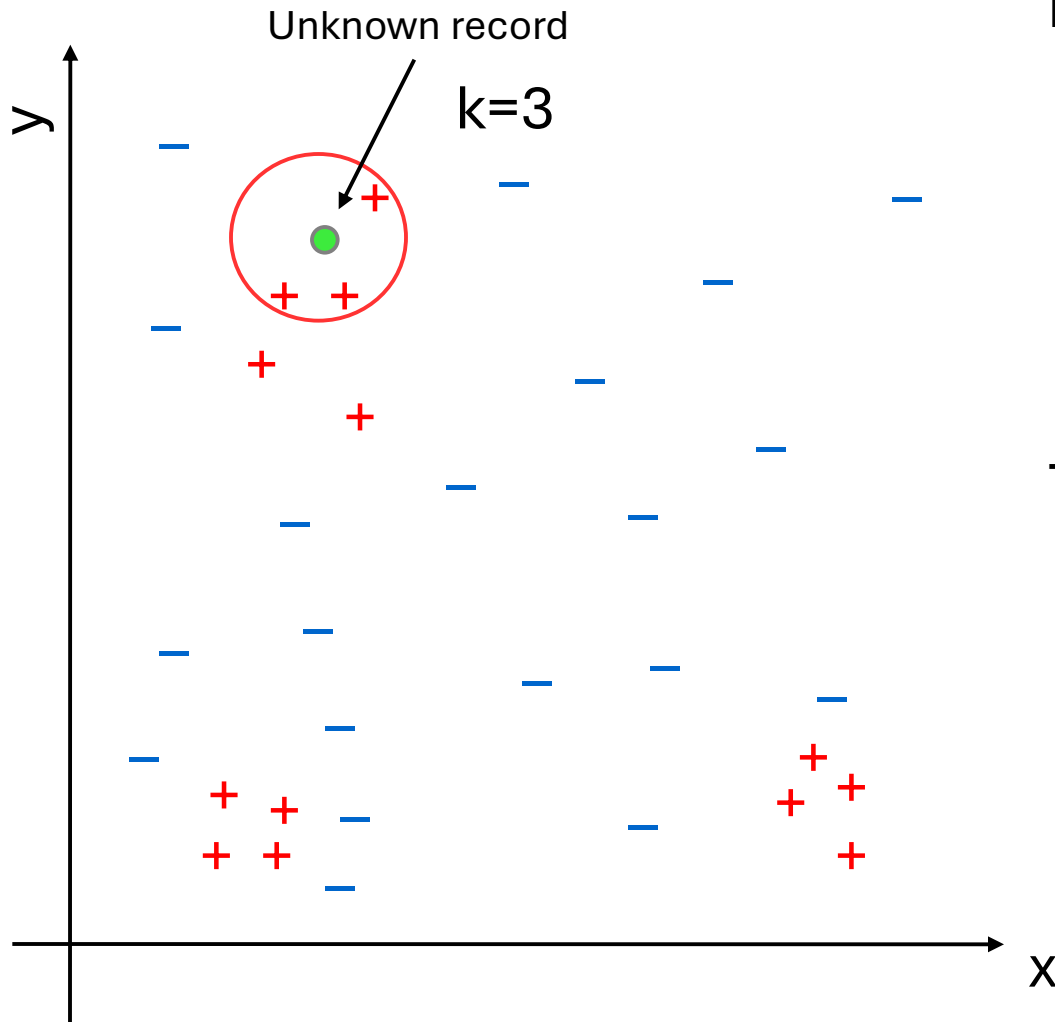
- Rule-Based Classifier
- **Nearest Neighbor Classifier**
- Naive Bayes Classifier

Nearest-Neighbor Classifiers

- Basic idea:
 - If it walks like a duck, quacks like a duck, then it's probably a duck



Nearest-Neighbor Classifiers



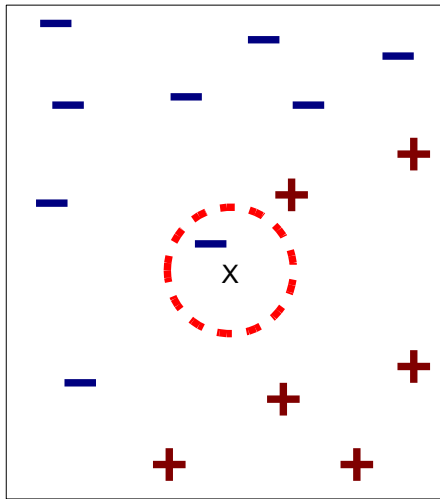
Requires three things

- The set of stored records.
- Distance Metric to compute the distance between records.
- The value of k , the number of nearest neighbors to retrieve.

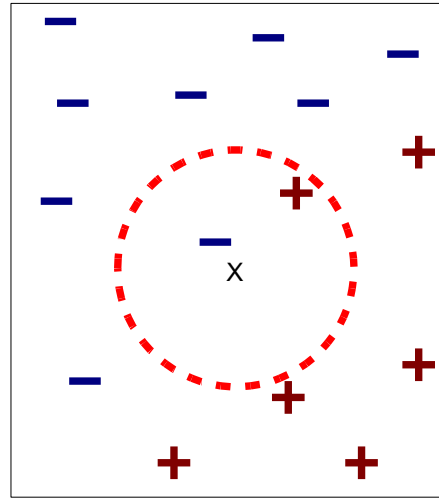
To classify an unknown record:

- Compute distance to other training records.
- Identify k nearest neighbors.
- Use class labels of nearest neighbors to determine the class label of an unknown record (e.g., by taking majority vote).

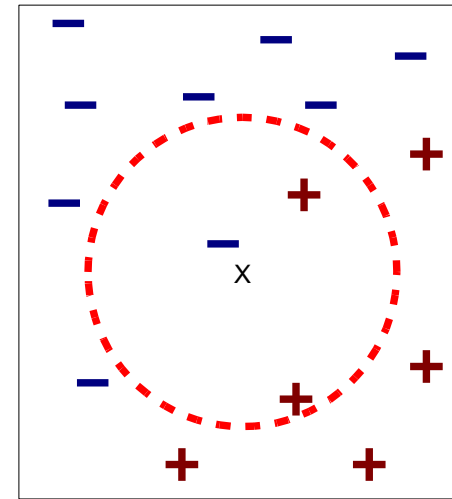
Definition of Nearest Neighbor



(a) 1-nearest neighbor



(b) 2-nearest neighbor



(c) 3-nearest neighbor

- k -nearest neighbors of a record x are data points with the k smallest distances to x .
- k is a hyperparameter.
- Odd numbers are preferable for k .

Distance Computation for Nearest-Neighbor Classification

- Compute distance between two points:
 - Typically uses Euclidean distance

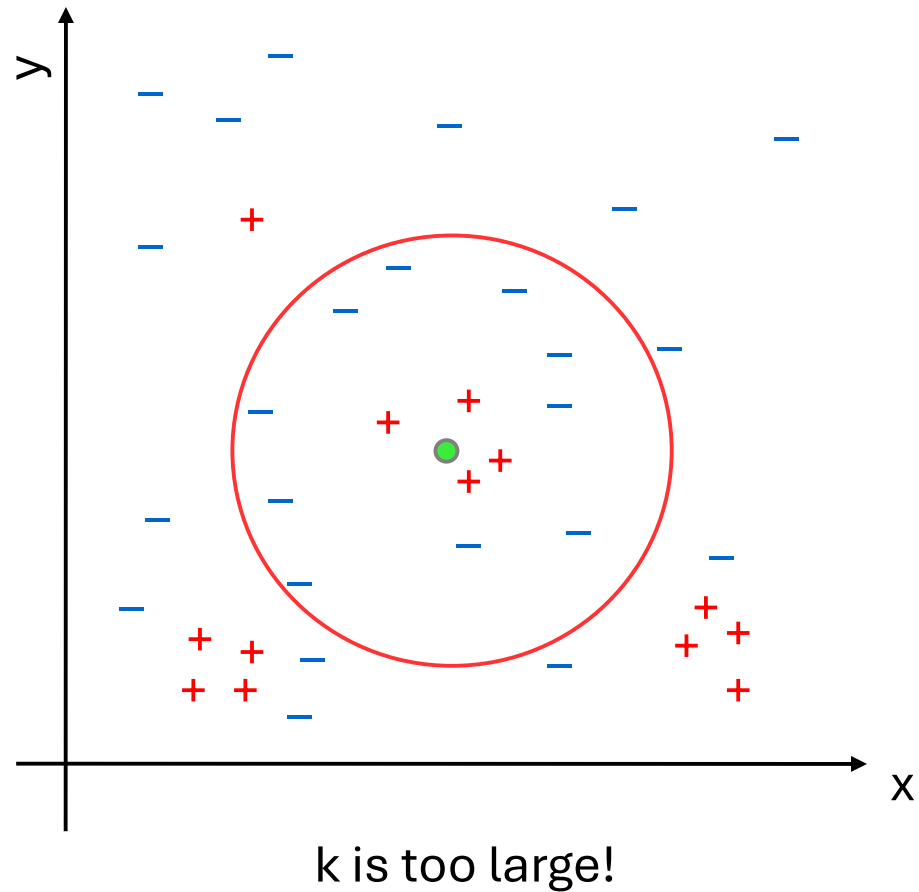
$$d(\mathbf{p}, \mathbf{q}) = \sqrt{\sum_i (p_i - q_i)^2}$$

Note: This means that the data needs to be **scaled**!

- Determine the class from nearest neighbor list. Options
 - a. Take the majority vote of class labels among the k-nearest neighbors.
 - b. Weigh the vote according to distance (e.g., weight factor $w = 1/d^2$).

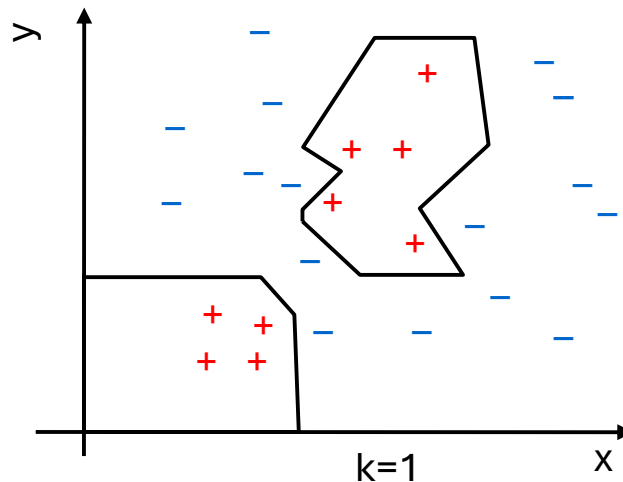
Choosing k

- If k is too small, sensitive to noise points
- If k is too large, neighborhood may include points from other classes



Advantages and Disadvantages

Advantage: Can create arbitrary non-linear decision boundaries.



Disadvantages: k-NN classifiers are lazy learners

- It does not build models explicitly (unlike eager learners such as decision trees).
- Needs to store all the training data.
- Classifying unknown records are relatively expensive (find the k-nearest neighbors). Space partitioning data structures like k-d trees can help.



Topics

- Rule-Based Classifier
- Nearest Neighbor Classifier
- **Naive Bayes Classifier**

Bayes' Rule

- The product rule gives us two ways to factor a joint distribution:

$$P(x, y) = P(x|y)P(y) = P(y|x)P(x)$$

- Therefore,

Posterior Prob.

Prior Prob.

$$P(y|x) = \frac{P(x|y) P(x)}{P(y)}$$

- Why is this useful?
 - Can get diagnostic probability $P(\text{cavity} | \text{toothache})$ from causal probability $P(\text{toothache} | \text{cavity})$
 - We can update our beliefs based on evidence.
 - Important tool for probabilistic inference .

Example of Bayes Theorem

- A doctor knows that meningitis causes a stiff neck 50% of the time

$$P(s|m) = .5$$

- The probability of any patient having meningitis is

$$P(m) = 1/50,000 = \mathbf{0.00002}$$

- The probability of any patient having stiff neck is

$$P(s) = 1/20 = 0.05$$

- If a patient has stiff neck, what's the probability he/she has meningitis?

$$P(m | s) = \frac{P(s | m) P(m)}{P(s)} = \frac{.5 \times 0.00002}{0.05} = \mathbf{0.0002}$$

Increases the probability by x10!

Bayesian Classification Rule

- Consider each attribute and class label as a random variable X_i taking the value x_i and Y taking y .
- Classification problem: Given a record with attributes $(x_1, x_2, \dots, x_n)$ predict class y .
- This can be done by finding the most likely class that has the largest

$$\operatorname{argmax}_y P(y | x_1, x_2, \dots, x_n)$$

- This classification rule is guaranteed **optimal** for the accuracy measure!

Bayesian Classifiers

- Compute the posterior probability $P(y | x_1, x_2, \dots, x_n)$ for all values of y using the Bayes theorem

$$\operatorname{argmax}_y P(y | x_1, x_2, \dots, x_n) = \operatorname{argmax}_C \frac{P(x_1, x_2, \dots, x_n | y) P(y)}{P(x_1, x_2, \dots, x_n)}$$

- This is equivalent to choosing value of C that maximizes
$$\operatorname{argmax}_y P(x_1, x_2, \dots, x_n | y) P(y)$$

This is a constant!
We don't need it for the max.

- Estimating the probability distribution $\mathbf{P}(Y)$ is easy, but how do we estimate

$$\mathbf{P}(X_1, X_2, \dots, X_n | Y)?$$

Unfortunately, the table for this probability distribution is very large and can only be estimated for a small number of attributes n .

Approximation of Bayesian Classifiers

- Decision trees use
$$\operatorname{argmax}_y P(y \mid \text{leafNodeMatching}(x_1, x_2, \dots, x_n))$$
- Rule-based classifiers use
$$\operatorname{argmax}_y P(y \mid \text{rulesMatching}(x_1, x_2, \dots, x_n))$$
- K-NN classifiers use
$$\operatorname{argmax}_y P(y \mid \text{neighborhood}(x_1, x_2, \dots, x_n))$$
- ANN classifiers with a final Softmax layer use
$$\operatorname{argmax}_y P(y \mid \text{activationBeforeSoftmaxLayer}(x_1, x_2, \dots, x_n))$$

Naïve Bayes Classifier

Approximates a Bayes Classifier by assuming independence among attributes X given the class. Now we can factor the probability distribution into the product of a few independent probabilities.

$$P(x_1, x_2, \dots, x_n | y) = P(x_1 | y) P(x_2 | y) \dots P(x_n | y) = \prod_i P(x_i | y)$$

We can estimate $P(x_i | y)$ for all x_i and y .

A new observation is classified as y such that:

$$\operatorname{argmax}_y P(y) \prod_i P(x_i | y)$$

How to Estimate Probabilities from Data?

Nominal Features

- Use the maximum likelihood estimate for probabilities.

- Class: $P(y) = \frac{N_y}{N}$
e.g., $P(Y = \text{No}) = 7/10$,
 $P(Y = \text{Yes}) = 3/10$

- For discrete attributes:

$$P(x_i | y) = \frac{N_{x_i \wedge y}}{N_y}$$

where $N_{x_i \wedge y}$ is number of instances having attribute x_i and belongs to class y .

e.g.,

$$P(\text{Status}=\text{Married} \mid y = \text{No}) = 4/7$$

$$P(\text{Refund}=\text{Yes} \mid y = \text{Yes}) = 0$$

<i>Tid</i>	Refund	Marital Status	Taxable Income	Class
1	Yes	Single	125K	No
2	No	Married	100K	No
3	No	Single	70K	No
4	Yes	Married	120K	No
5	No	Divorced	95K	Yes
6	No	Married	60K	No
7	Yes	Divorced	220K	No
8	No	Single	85K	Yes
9	No	Married	75K	No
10	No	Single	90K	Yes

How to Estimate Probabilities from Data?

Continuous Features

Several options:

- Discretize the range into bins
 - one binary variable per bin (one-hot encoding).
 - violates the independence assumption.
- Two-way split: $(x_i < v)$ or $(x_i > v)$
 - Encode with one binary variable.
- Probability density estimation.
 - Assume the attribute follows a normal distribution.
 - Use data to estimate the parameters of the distribution (e.g., mean and standard deviation).
 - Once the probability distribution is known, we can use it to estimate the conditional probability $P(x_i | y)$.
 - Most implementations will do this automatically. This is called a Gaussian Naïve Bayes Classifier.

Example of Naïve Bayes Classifier

Given a Test Record what is the most likely class?

$x = (\text{Refund}=\text{No}, \text{Married}, \text{Income} = 120K)$

naive Bayes Classifier:

$P(\text{Refund}=\text{Yes}|\text{No}) = 3/7$
 $P(\text{Refund}=\text{No}|\text{No}) = 4/7$
 $P(\text{Refund}=\text{Yes}|\text{Yes}) = 0$
 $P(\text{Refund}=\text{No}|\text{Yes}) = 1$
 $P(\text{Marital Status}=\text{Single}|\text{No}) = 2/7$
 $P(\text{Marital Status}=\text{Divorced}|\text{No}) = 1/7$
 $P(\text{Marital Status}=\text{Married}|\text{No}) = 4/7$
 $P(\text{Marital Status}=\text{Single}|\text{Yes}) = 2/7$
 $P(\text{Marital Status}=\text{Divorced}|\text{Yes}) = 1/7$
 $P(\text{Marital Status}=\text{Married}|\text{Yes}) = 0$

For taxable income:

If class=No: sample mean=110
 sample variance=2975

If class=Yes: sample mean=90
 sample variance=25

$$\begin{aligned} P(x|\text{Class}=\text{No}) &= P(\text{Refund}=\text{No}|\text{Class}=\text{No}) \\ &\quad * P(\text{Married}|\text{Class}=\text{No}) \\ &\quad * P(\text{Income}=120K|\text{Class}=\text{No}) \\ &= 4/7 * 4/7 * 0.0072 = 0.0024 \end{aligned}$$

$$\begin{aligned} P(x|\text{Class}=\text{Yes}) &= P(\text{Refund}=\text{No}|\text{Class}=\text{Yes}) \\ &\quad * P(\text{Married}|\text{Class}=\text{Yes}) \\ &\quad * P(\text{Income}=120K|\text{Class}=\text{Yes}) \\ &= 1 * 0 * 1.2 * 10^{-9} = 0 \end{aligned}$$

0s are an issue!

$$P(\text{No}|x) = P(x|\text{No})P(\text{No}) > P(\text{Yes}|x) = P(x|\text{Yes})P(\text{Yes})$$

Predicted Class is No

Naïve Bayes Classifier: Dealing With Low Counts

Probability estimation:

Original:
$$P(x_i | y) = \frac{N_{x_i \wedge y}}{N_y}$$

Issue: If one of the conditional probabilities is zero, then the entire expression becomes zero.

Laplace:
$$P(x_i | y) = \frac{N_{x_i \wedge y} + 1}{N_y + c}$$

c : number of classes
 m : parameter

m-estimate:
$$P(x_i | y) = \frac{N_{x_i \wedge y} + mP(y)}{N_y + m}$$



Summary of Naïve Bayes Classifiers

- **Robust to outliers** and isolated noise points since it is not based on distances.
- **Can handle missing value** during prediction: Ignore the attribute during probability estimate calculations.
- **Robust to irrelevant attributes:** Features are estimated independently. Irrelevant features will produce a likelihood that is a uniform distribution given the class.
- **Independence assumption** may not hold for some attributes
 - Typically, the classifiers still work well when the assumption is slightly violated.
 - You can remove highly correlated attributes.
 - Use other techniques such as Bayesian Belief Networks (BBN) that explicitly model dependence.